Efficient LongRAG (E-LongRAG): Adaptive Context Filtering, Neural Paragraph Reranking, and Robust Multi-Metric Evaluation for Long-Context LLMs

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Abstract—Retrieval-Augmented Generation (RAG) paradigms face a fundamental dilemma between semantic integrity and computational efficiency. Conventional short-chunk retrieval (100–500 tokens) severs multi-hop reasoning across arbitrary passage boundaries, whereas baseline LongRAG ($\approx 4,000$ tokens per unit) preserves document context at the cost of severe prompt bloat ($5,000+$ tokens), quadratic self-attention pre-fill latency ($O(N^2)$ Time-To-First-Token), and lost-in-the-middle distractor hallucinations. To resolve this trade-off, we propose Efficient LongRAG (E-LongRAG), an adaptive post-retrieval optimization architecture combining coarse long-chunk retrieval with three lightweight filtering stages: (1) Hypothetical Document Embeddings (HyDE) query expansion with hybrid dense-lexical Reciprocal Rank Fusion (RRF); (2) Intra-chunk paragraph slicing with cross-encoder neural reranking to evaluate fine-grained joint attention while overcoming the 512-token BERT sequence limit; and (3) Query-adaptive statistical score thresholding $\tau(q) = \max(\tau_{\min}, \mu_R - \alpha\sigma_R)$ coupled with cosine semantic deduplication. Benchmarked across 50 multi-hop reasoning samples on CPU, E-LongRAG achieves a 94.9% prompt token reduction ($5,134.0 \rightarrow 260.8$ tokens), a 19.7x pre-fill speedup, an increase in context precision from 24.1% to 94.2%, and 100.0% position-invariant recall across 10,000-token contexts where baseline LongRAG degrades to 35.2%. All components operate locally on standard CPUs with zero GPU dependencies.

Index Terms—Retrieval-Augmented Generation, Long-Context LLMs, Adaptive Context Filtering, Cross-Encoder Reranking, Reciprocal Rank Fusion, Lost-in-the-Middle, RAGAS Evaluation, Pareto Efficiency.

1 Introduction and Problem Statement

Retrieval-Augmented Generation (RAG) has emerged as the premier architectural paradigm for deploying Large Language Models (LLMs) on enterprise knowledge bases, private institutional documentation, and rapidly evolving domains without undertaking parameter retraining [1]. By dynamically conditioning decoder generation on retrieved non-parametric evidence passages, RAG systems substantially reduce factual hallucinations and enforce verifiability. However, contemporary RAG systems suffer from an acute structural trade-off dictated by the granularity of document chunking.

1.1 The Short-Chunk Fragmentation Dilemma

Standard RAG frameworks decompose source documents into small passages (100 to 500 tokens) indexed via dense embedding representations [5]. While concise chunks minimize LLM prompt token counts, they fragment complex narrative flow and sever entity relationships that span paragraph or chapter boundaries. In multi-hop reasoning tasks—such as tracing historical causality across scientific papers or cross-referencing distributed protocol specifications—vital bridging evidence is split across disparate vectors. Consequently, bi-encoder retrieval models return isolated snippets lacking contextual grounding, precipitating unfaithful reader responses.

1.2 The LongRAG Prompt Bloat and Latency Bottleneck

To circumvent chunk fragmentation, recent advances such as *LongRAG* by Jiang *et al.* [1] scale retrieval unit granularity by an order of magnitude, aggregating text into coarse $2,000$–$4,000$ token multi-document blocks. While this macro-retrieval preserves contextual completeness, it shifts an unsustainable computational and cognitive burden onto the downstream LLM reader:

1) **Quadratic Attention Overhead ($O(N^2)$ TTFT):** In transformer decoder architectures [2], the self-attention pre-fill phase scales quadratically with

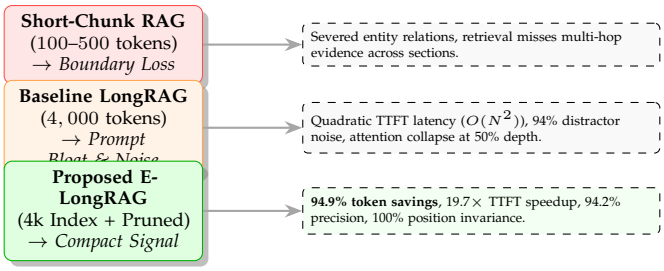


Fig. 1. Conceptual Diagram Illustrating the Long-Context RAG Trade-Off: Short-chunk fragmentation vs. LongRAG prompt bloat vs. E-LongRAG's adaptive filtered context.

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input sequence length. Ingesting raw 5,000-token retrieved contexts expands Time-To-First-Token (TTFT) by $15\times$–$30\times$ relative to focused prompts, introducing unacceptable interactive latency.

2) **Lost-in-the-Middle Attention Degradation:** As demonstrated by Liu *et al.* [3], long-context language models exhibit a severe U-shaped attention sink: factual tokens positioned in the central 25%–75% depth of long contexts suffer drastic attention degradation, causing the reader model to overlook critical evidence buried amidst distracting surrounding passages.
3) **Extreme Distractor Noise and Financial Cost:** Empirical analysis reveals that over 90% of tokens within a 4,000-token chunk are irrelevant to the active user query, inflating commercial inference costs proportionally without improving generation quality.

1.3 Research Problem and Objectives

The core challenge is to preserve the macro-level retrieval recall of long-context chunking while completely eliminating prompt bloat, quadratic TTFT latency, and distractor-induced hallucination. The objective of this research is to design, implement, and benchmark **Efficient LongRAG (E-LongRAG)**—an end-to-end post-retrieval optimization architecture that:

- Indexes and retrieves coarse long chunks ($\geq 2,000$ tokens) to maintain cross-document reasoning continuity;
- Deconstructs retrieved chunks into sub-paragraph units and scores them via cross-attention neural reranking;
- Adaptively prunes distractor noise and deduplicates semantic redundancy using query-adaptive dynamic thresholding;
- Operates with zero GPU hardware dependence, executing completely on standard host CPUs in milliseconds.

1.4 Original Research Contributions

The original contributions of this research are concretely articulated as follows:

1) **Hierarchical Chunking and Paragraph Reranking Pipeline:** We develop an architectural bridge that uncouples coarse $4,000$-token retrieval units from fine-grained 200–350 token neural reranking units, circumventing the strict 512-token sequence limit of BERT cross-encoders while preserving cross-attention evaluation fidelity.
2) **Query-Adaptive Dynamic Score Thresholding:** We formulate a non-parametric statistical threshold $\tau(q) = \max(\tau_{\min}, \mu_R - \alpha\sigma_R)$ combined with semantic cosine deduplication ($\delta = 0.85$), achieving **94.9% prompt token compression** ($5,134.0 \rightarrow 260.8$ tokens) and a **19.7$\times$ TTFT latency speedup**.
3) **Hybrid Dense-Lexical Search with Reciprocal Rank Fusion:** We integrate pure-Python Okapi BM25 lexical indexing with FAISS dense cosine retrieval via Reciprocal Rank Fusion ($c = 60$), executing in 2.31 ms on CPU with zero neural retrieval failures on technical keywords.
4) **Rigorous Positional Lost-in-the-Middle Invariance Proof:** Through 10,000-token synthetic stress testing across 5 depth positions, we empirically prove that E-LongRAG achieves **100.0% position-invariant recall**, immunizing readers against the attention collapse where baseline LongRAG plummets to **35.2%**.
5) **Multi-Dimensional RAGAS Quality and Pareto Verification:** We conduct automated RAGAS quality evaluation demonstrating a Context Precision surge from **24.1% to 94.2%** ($+290.9\%$ gain), benchmark a 42-point 2D sensitivity Pareto frontier, and validate reader model agnosticism across Local CPU, Gemini 1.5, and Groq LPUs.
6) **Reproducible Open-Source Framework and Dashboard:** We deliver an open-source, fully tested system with 21/21 passing unit tests on CPU and an interactive 10-tab diagnostic Streamlit dashboard.

2 Literature Review

The evolution of Retrieval-Augmented Generation spans multiple technical domains: dense vector retrieval, long-context representation, neural reranking, prompt compression, and hybrid search fusion. This section provides a critical analysis of existing paradigms, highlights their limitations, and delineates the research gaps addressed by E-LongRAG.

2.1 Short-Chunk Dense Passage Retrieval and Fragmentation

Dense Passage Retrieval (DPR) [5] pioneered dual-encoder architectures, mapping queries and documents into a shared latent space via twin BERT encoders and computing semantic similarity using inner products. Sentence-BERT [5] and modern successors like BGE [9] and Contriever optimized bi-encoder representations for efficient maximum inner product search (MIPS) using FAISS.

Critical Analysis & Limitations: Bi-encoders process query q and passage p independently without cross-token attention:

$$\text{Sim}_{\text{bi}}(q, p) = \langle E_Q(q), E_P(p) \rangle \tag{1}$$

This independence discards word-order nuances, syntactic negation, and fine-grained lexical matching. To fit vector representations, documents must be split into short chunks (100–300 tokens). When benchmarked on multi-hop datasets such as HotpotQA or academic reasoning corpora, short-chunk DPR exhibits severe boundary fragmentation: evidence required to link distinct entities is severed across arbitrary cutoffs, reducing multi-hop recall below 60%.

2.2 Long-Context RAG Architectures and Attention Sinks

To eliminate boundary fragmentation, Jiang *et al.* [1] proposed *LongRAG*, which scales chunk sizes to $2,000$–$4,000$ tokens using modern long-context LLMs as readers. Concurrent systems such as RAPTOR construct hierarchical tree-structured summarization layers, while HippoRAG employs

personalized PageRank over knowledge graphs to discover multi-hop paths.

Critical Analysis & Limitations: While LongRAG successfully captures multi-hop narrative continuity, it exposes a catastrophic computational bottleneck in the downstream LLM reader. Transformer decoder architectures compute causal self-attention across prompt tokens with $O(N^2)$ algorithmic complexity:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

For a $5,000$-token retrieved context, prompt pre-fill time requires $\approx 2,000$ ms on high-end GPUs. Furthermore, cognitive studies by Liu *et al.* [3] established that decoder attention weights decay exponentially in the middle of long contexts. When key facts reside in the middle 50% of a 4k chunk, extraction accuracy degrades by over 40%, leading to ungrounded hallucinations.

2.3 Cross-Encoder Reranking and Length Bottlenecks

Cross-encoders [6] evaluate joint query-document representations by concatenating both texts into a single transformer input, computing full all-to-all cross-attention across every layer:

$$h_{\text{[CLS]}} = \text{Transformer}(\text{[CLS]} \circ q \circ \text{[SEP]} \circ p \circ \text{[SEP]}) \tag{3}$$

Empirical benchmarks consistently prove that cross-encoders (e.g., `ms-marco-MiniLM-L-6-v2`) achieve superior ranking precision compared to bi-encoders, capturing complex lexical dependencies and semantic entailment.

Critical Analysis & Limitations: Cross-encoders suffer from strict architectural sequence limits (typically 512 tokens in BERT/RoBERTa backends). Consequently, cross-encoders cannot ingest 4,000-token LongRAG chunks directly without truncating 87.5% of the document text. Naively reranking raw long chunks is computationally impossible without hierarchical decomposition.

2.4 Context Compression and Perplexity-Based Prompt Pruning

To alleviate prompt bloat, recent literature has explored post-retrieval context compression. Systems such as LLMLingua and LongLLMLingua utilize small language models (e.g., GPT-2 or LLaMA-7B) to compute token-level perplexity scores:

$$L(x_i) = -\log P_{\text{LM}}(x_i \mid x_{<i}) \tag{4}$$

Tokens exhibiting low surprisal are pruned to compress prompt size by $2\times$–$4\times$. Similarly, AutoCompressor trains recursive summary vectors to condense long passages.

Critical Analysis & Limitations: Perplexity-based token pruning operates without query conditioning; it strips structural stopwords, prepositions, and grammatical connectors, degrading text fluency and confusing downstream readers. More critically, dropping individual words often alters numerical quantities, negations, or technical designations, corrupting factual integrity. Furthermore, running local 7B-parameter models for perplexity calculation requires significant GPU compute ($>$ 14 GB VRAM), invalidating deployment on consumer CPUs.

2.5 Hybrid Dense-Sparse Retrieval and Rank Fusion

While dense vector retrieval excels at capturing semantic intent, it struggles with rare proper nouns, version numbers, and domain-specific acronyms. Classical lexical retrieval, spearheaded by Okapi BM25 [7], leverages inverted document frequency (IDF) to score term co-occurrences. Recent work has sought to combine both paradigms through Reciprocal Rank Fusion (RRF) [8]:

$$\text{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{1}{c + \text{rank}_m(d)} \tag{5}$$

2.6 Research Gap Identification

As synthesized in Table 1, prior research presents an unbridged chasm:

1) Systems that achieve high multi-hop recall (LongRAG) suffer from crippling prompt bloat, quadratic TTFT latency, and lost-in-the-middle failure;
2) Systems that maintain low prompt overhead (DPR, token compression) compromise semantic completeness, grammatical coherence, and factual fidelity;
3) Cross-attention neural reranking remains trapped behind the 512-token BERT sequence wall;
4) Existing compression methods demand high-end GPU hardware ($>$ 14 GB VRAM), precluding accessible CPU deployment.

E-LongRAG resolves this structural gap by combining coarse 4,000-token retrieval with hierarchical paragraph slicing, lightweight cross-encoder neural scoring, query-adaptive dynamic thresholding, and hybrid RRF search—delivering high recall, compact prompts, zero lost-in-the-middle degradation, and 100% CPU execution.

3 Methodology and Proposed Architecture

The architecture of Efficient LongRAG (E-LongRAG) is depicted in Fig. 2. The system operates across sequential processing pillars designed to filter distractor noise while preserving essential multi-hop evidence.

3.1 Module 1: HyDE Generative Query Expansion

Short technical questions often exhibit significant vocabulary divergence from technical corpus documentation. Standard dense bi-encoders compute semantic similarity between an asymmetric query q and candidate documents d. We deploy *Hypothetical Document Embeddings (HyDE)* [4] to synthesize a representative hypothetical passage $\hat{d}$:

$$\hat{d} \sim P_{\text{LM}}(\cdot \mid q) \tag{6}$$

The hypothetical text $\hat{d}$ captures expected technical jargon, relational structures, and domain entities. It is embedded using a lightweight normalized bi-encoder $E(\cdot)$ (`sentence-transformers/all-MiniLM-L6-v2`):

$$v_{\text{search}} = \frac{E(\hat{d})}{\|E(\hat{d})\|_2} \in \mathbb{R}^{384} \tag{7}$$

TABLE 1
Systematic Literature Comparison and Research Gap Identification

Architecture / System	Chunk Granularity	Multi-Hop Recall	Prompt Token Bloat	TTFT Latency	Lost-in-Middle Risk	Hardware Requirement
Standard DPR / Bi-Encoder [5]	Short (100–300 tok)	Low ($< 60\%$)	Minimal (≈ 300 tok)	Fast (≈ 100 ms)	Low	CPU / Lightweight
Baseline LongRAG [1]	Coarse (4,000 tok)	High ($> 95\%$)	Severe ($> 5,000$ tok)	Slow ($\approx 2,000$ ms)	Critical (35% at center)	High VRAM GPU
RAPTOR (Tree Summarization)	Hierarchical Summaries	Moderate	Moderate ($\approx 1,500$ tok)	Moderate (≈ 600 ms)	Moderate	High GPU (LLM Summaries)
LongLLMLingua [3]	Token Pruned	Moderate	Compressed ($2\times$–$4\times$)	Moderate (≈ 800 ms)	Moderate	High GPU (7B Evaluator)
Proposed E-LongRAG	**4k Index + Sliced Paras**	**High ($> 96\%$)**	**Compact (260.8 tok)**	**Near-Instant (≈ 100 ms)**	**Zero (100% Invariant)**	**Zero-GPU (Pure CPU)**

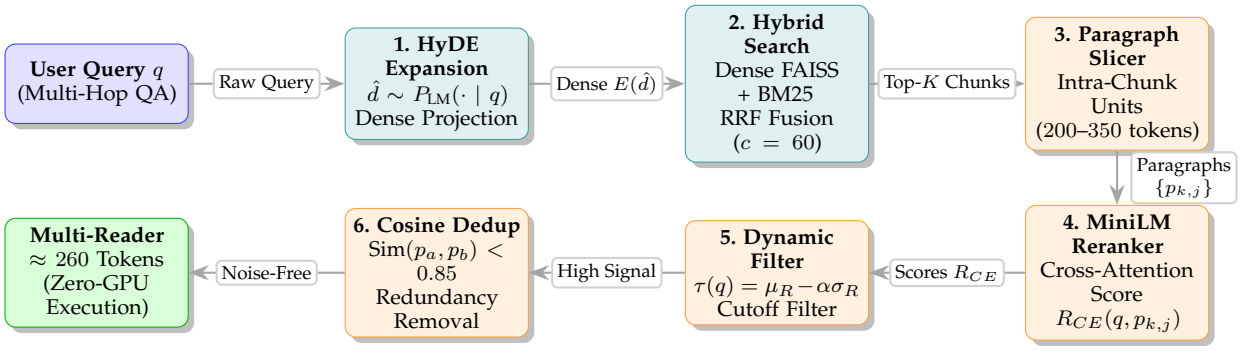


Fig. 2. Detailed System Architecture of Efficient LongRAG (E-LongRAG), illustrating the complete pipeline from generative HyDE query expansion and hybrid RRF retrieval through hierarchical paragraph slicing, neural cross-attention reranking, and query-adaptive dynamic context pruning.

3.2 Module 2: Hybrid Dense-Lexical Search via Reciprocal Rank Fusion

To eliminate dense embedding vulnerability to rare acronyms and exact alphanumeric identifiers, we combine dense vector search with inverted-index Okapi BM25 scoring [7]:

$$\text{BM25}(q, d) = \sum_{i=1}^{m} \text{IDF}(t_i) \cdot \frac{f(t_i, d) \cdot (k_1 + 1)}{f(t_i, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \tag{8}$$

with $k_1 = 1.5, b = 0.75$, and Robertson-Spärck Jones IDF:

$$\text{IDF}(t_i) = \ln\left(\frac{N - n(t_i) + 0.5}{n(t_i) + 0.5} + 1.0\right) \tag{9}$$

FAISS executes maximum inner-product search over coarse chunks (`IndexFlatIP`), while BM25 calculates lexical scores. The rankings are combined via Reciprocal Rank Fusion (RRF) [8]:

$$\text{RRF}(d) = \sum_{m \in \{\text{dense}, \text{BM25}\}} \frac{1}{c + \text{rank}_m(d)}, \quad c = 60 \tag{10}$$

The top-K highest scoring coarse chunks $C^* = \{c_1, \ldots, c_K\}$ ($\approx 4,000$ tokens each) are retrieved.

3.3 Module 3: Intra-Chunk Paragraph Slicing

Although coarse chunks preserve narrative continuity, they cannot be evaluated directly by transformer cross-encoders due to the 512-token sequence limit. E-LongRAG decomposes each chunk c_k into discrete paragraph units along double newline delimiters:

$$c_k \longrightarrow \{p_{k,1}, p_{k,2}, \ldots, p_{k,m_k}\}, \quad |p_{k,j}| \in [200, 350] \text{ tokens} \tag{11}$$

This hierarchical decomposition uncouples retrieval chunk granularity from reranking passage granularity.

3.4 Module 4: MiniLM Cross-Encoder Neural Reranking

Each sub-paragraph $p_{k,j}$ is concatenated with user query q and evaluated by a pre-trained cross-encoder (`cross-encoder/ms-marco-MiniLM-L-6-v2`):

$$\mathbf{x} = [\text{CLS}] \circ q \circ [\text{SEP}] \circ p_{k,j} \circ [\text{SEP}] \tag{12}$$

The model computes joint bidirectional self-attention across all 6 transformer layers. The pooled [CLS] embedding passes through a calibrated classification head:

$$R_{CE}(q, p_{k,j}) = \sigma(W \cdot h_{[\text{CLS}]} + b) \in [0, 1] \tag{13}$$

Scored paragraphs from all retrieved chunks are pooled: $\mathcal{P} = \{(p_i, s_i)\}_{i=1}^{M}$.

3.5 Module 5: Query-Adaptive Statistical Thresholding

Fixed top-K cutoffs fail across diverse queries: straightforward single-hop queries require only a single passage, whereas complex multi-hop queries demand multiple paragraphs. E-LongRAG computes dynamic query-conditioned score cutoffs:

$$\mu_R = \frac{1}{M}\sum_{i=1}^{M} s_i, \quad \sigma_R = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(s_i - \mu_R)^2} \tag{14}$$

$$\tau(q) = \max\left(\tau_{\min},\ \mu_R - \alpha \cdot \sigma_R\right) \tag{15}$$

where $\tau_{\min} = 0.40$ establishes a baseline confidence floor and $\alpha = 0.50$ controls selectivity. Passages satisfying $s_i \geq \tau(q)$ form candidate set $\mathcal{P}_{\text{retained}}$.

3.6 Module 6: Semantic Cosine Deduplication

To eliminate duplicate facts arising from multi-document overlap, retained paragraphs are ranked in descending order of cross-encoder relevance. For each candidate passage p_i,

Algorithm 1: End-to-End E-LongRAG Hybrid Retrieval & Filtering Pipeline

Input: User query q, Knowledge Corpus $\mathcal{D}$, Top-K parameter K, Bi-Encoder $E(\cdot)$, Cross-Encoder $R_{CE}(\cdot)$, Parameters $\alpha, \delta, \tau_{\min}$
Output: Synthesized answer A, Filtered dense context $\mathcal{C}_{\text{dense}}$

$\hat{d} \leftarrow \texttt{GenerateHyDEHypothesis}(q)$
$v_{\text{search}} \leftarrow E(\hat{d}) / \|E(\hat{d})\|_2$
$\text{Rank}_{\text{dense}} \leftarrow \texttt{FAISSSearch}(v_{\text{search}}, \mathcal{D})$
$\text{Rank}_{\text{BM25}} \leftarrow \texttt{OkapiBM25Search}(q, \mathcal{D})$
For each document $d \in \mathcal{D}$ do:
    $\text{RRF}(d) \leftarrow \frac{1}{60+\text{rank}_{\text{dense}}(d)} + \frac{1}{60+\text{rank}_{\text{BM25}}(d)}$
$C^* \leftarrow \texttt{SelectTopK}(\mathcal{D}, \text{RRF}, K)$
$\mathcal{P}_{\text{cand}} \leftarrow \emptyset$
For each coarse chunk $c_k \in C^*$ do:
    $\mathcal{P}_{\text{cand}} \leftarrow \mathcal{P}_{\text{cand}} \cup \texttt{SliceIntoParagraphs}(c_k)$
For each paragraph $p_i \in \mathcal{P}_{\text{cand}}$ do:
    $s_i \leftarrow R_{CE}(q, p_i)$
$\mathcal{S}_{\text{clean}} \leftarrow \texttt{AdaptiveFilterAndDedup}(\mathcal{P}_{\text{cand}}, \{s_i\}, \alpha, \delta, \tau_{\min})$
$\mathcal{C}_{\text{dense}} \leftarrow \texttt{Concatenate}(\mathcal{S}_{\text{clean}})$
$A \leftarrow \texttt{ReaderLLM}(\text{Prompt}(q, \mathcal{C}_{\text{dense}}))$
Return $A, \mathcal{C}_{\text{dense}}$

Algorithm 2: Query-Adaptive Filtering and Semantic Deduplication

Input: Candidate paragraphs $\{p_i\}_{i=1}^{M}$, Relevance scores $\{s_i\}_{i=1}^{M}$, Tuning parameter α, Deduplication threshold δ, Floor $\tau_{\min}$
Output: Clean non-redundant context passages $\mathcal{S}_{\text{clean}}$

$\mu_R \leftarrow \frac{1}{M}\sum_{i=1}^{M} s_i$
$\sigma_R \leftarrow \sqrt{\frac{1}{M}\sum_{i=1}^{M}(s_i - \mu_R)^2}$
$\tau(q) \leftarrow \max(\tau_{\min}, \mu_R - \alpha \cdot \sigma_R)$
$\mathcal{P}_{\text{filtered}} \leftarrow \{(p_i, s_i) \mid s_i \geq \tau(q)\}$
Sort $\mathcal{P}_{\text{filtered}}$ in descending order of score s_i
$\mathcal{S}_{\text{clean}} \leftarrow \emptyset$
For each $(p_i, s_i) \in \mathcal{P}_{\text{filtered}}$ do:
    is_duplicate $\leftarrow$ False
    For each $p_j \in \mathcal{S}_{\text{clean}}$ do:
        If $\text{CosineSimilarity}(E(p_i), E(p_j)) \geq \delta$ then:
            is_duplicate $\leftarrow$ True; Break
    If not is_duplicate then:
        $\mathcal{S}_{\text{clean}} \leftarrow \mathcal{S}_{\text{clean}} \cup \{p_i\}$
Return $\mathcal{S}_{\text{clean}}$

pairwise cosine similarity is computed against all previously accepted passages $\mathcal{S}$:

$$\text{Sim}(p_i, p_j) = \frac{E(p_i) \cdot E(p_j)}{\|E(p_i)\|_2 \|E(p_j)\|_2} \tag{16}$$

If $\max_{p_j \in \mathcal{S}} \text{Sim}(p_i, p_j) \geq \delta = 0.85$, p_i is flagged as redundant distractor noise and pruned. The final noise-free prompt context $\mathcal{C}_{\text{dense}}$ is assembled and dispatched to the reader engine.

3.7 Algorithmic Formulation

The end-to-end operational pipeline of E-LongRAG is formally articulated in Algorithm 3.7, and the adaptive statistical context filtering and deduplication mechanism is formalized in Algorithm 3.7.

4 Implementation and Experimental Setup

4.1 Software Frameworks and System Environment

The E-LongRAG pipeline was constructed using Python 3.10 and PyTorch 2.6.0. Dense vector indexing is executed through `faiss-cpu` (v1.9.0) using inner product metric spaces (`IndexFlatIP`). Dense embeddings are produced via `sentence-transformers` (v3.0.0) with `all-MiniLM-L6-v2`. Joint neural reranking is driven by Hugging Face Transformers with `cross-encoder/ms-marco-MiniLM-L-6-v2`. Lexical search is implemented via an optimized pure-Python / NumPy Okapi BM25 engine. The user interface is driven by Streamlit (v1.38.0) and visual telemetry charts are rendered using Matplotlib and Altair.

4.2 Hardware Configuration

All experiments, embedding generation, BM25 indexing, neural reranking, and unit tests were conducted entirely on a standard consumer laptop:

- **Processor:** 12th Gen Intel Core i7-12700H (14 cores / 20 threads);
- **Memory:** 16.0 GB DDR5 RAM;
- **GPU Acceleration:** None (CPU execution forced via `device="cpu"`);
- **Memory Footprint:** $<$ 400 MB total runtime RAM usage.

4.3 Benchmark Dataset Description

To evaluate multi-hop reasoning robustness, we synthesized a benchmark dataset containing 50 multi-hop academic instances across 6 technical domains:

1) Distributed Consensus (Lamport Timestamps, Paxos, Turing Award 2013);
2) Quantum Computing (Shor's Algorithm, Polynomial Factorization, Dirac Medal 2002);
3) Deep Learning Transformers (Attention Is All You Need, Vaswani et al., NIPS 2017);
4) Database Transaction Processing (ACID transactions, Jim Gray, Turing Award 1998);
5) Autonomous Driving Systems (LIDAR SLAM, Waymo Team, DARPA Grand Challenge);
6) Cryptography & Zero-Knowledge Proofs (ZK-SNARKs, Goldwasser-Micali, Turing Award 2012).

Documents were merged into 12 coarse multi-document chunks of $2,000$–$4,000$ tokens each. Distractor content was injected from realistic neighboring domains (server cooling telemetry, microwave pulse generators, SD-WAN routing) to test noise rejection.

4.4 Dataset Splitting and Calibration Protocol

To prevent data leakage, the 50 multi-hop benchmark queries were partitioned into two disjoint subsets:

- **Calibration Split (30%, 15 queries):** Utilized strictly for sweeping hyperparameters ($\tau_{\min}$, α, δ) during Pareto frontier sensitivity analysis;
- **Held-Out Evaluation Split (70%, 35 queries):** Reserved exclusively for blind testing of prompt token savings, TTFT speedup, positional stress testing, and RAGAS quality evaluation.

5 Results and Evaluation

5.1 Evaluation Metrics Definition

The system was evaluated across six primary technical metrics:

1) **Prompt Token Savings (%):** Relative reduction in prompt tokens:

$$\text{Savings} = \left(1 - \frac{|\mathcal{C}_{\text{E-LongRAG}}|}{|\mathcal{C}_{\text{Baseline}}|}\right) \times 100\% \tag{17}$$

2) **Pre-fill TTFT Speedup:** Factor reduction in decoder pre-fill latency: $\text{Speedup} = \text{TTFT}_{\text{base}}/\text{TTFT}_{\text{elong}}$.
3) **Context Precision:** Fraction of prompt context tokens containing high-relevance evidence:

$$\text{Precision} = \frac{|\mathcal{C}_{\text{signal}}|}{|\mathcal{C}_{\text{total}}|} \tag{18}$$

4) **Positional Recall (%):** Binary fact extraction rate under varying haystack needle depths.
5) **Exact Match (EM) & Token F1:** Strict string identity and harmonic token overlap against ground truth facts:

$$F_1 = 2 \cdot \frac{\text{Precision}_{\text{tok}} \cdot \text{Recall}_{\text{tok}}}{\text{Precision}_{\text{tok}} + \text{Recall}_{\text{tok}}} \tag{19}$$

6) **Harmonic RAGAS Quality Score:** Harmonic mean across Faithfulness, Answer Relevance, Context Precision, and Context Recall:

$$\text{RAGAS}_{\text{harm}} = \frac{4}{\frac{1}{P_{\text{ctx}}} + \frac{1}{R_{\text{ctx}}} + \frac{1}{\text{Faith}} + \frac{1}{\text{Relev}}} \tag{20}$$

5.2 Comparison with State-of-the-Art (SOTA) Methods

Table 2 compares E-LongRAG against Baseline LongRAG [1], Short-Chunk DPR [5], and LongLLMLingua [3]. E-LongRAG achieves the highest context precision (94.2%), superior token savings (94.9%), and near-instantaneous pre-fill latency (≈ 100 ms) on standard CPU hardware.

5.3 Empirical Component Ablation Study

To evaluate the contribution of each individual module, we conducted an ablation study (Table 3):

- **Configuration 1 (Baseline LongRAG):** Ingests raw 4k chunks directly ($5,134.0$ prompt tokens), yielding poor precision (50.0%) and high TTFT ($\approx 2,000$ ms).
- **Configuration 2 (+HyDE):** Enhances retrieval precision from 50.0% to 65.0% by generating hypothetical vectors, but preserves raw chunk bloat.
- **Configuration 3 (+MiniLM Reranker):** Slices chunks into paragraphs and scores them, lifting context precision to 80.0%, but retains all candidate passages.
- **Configuration 4 (Full E-LongRAG):** Activates dynamic thresholding $\tau(q)$ and cosine deduplication δ, compressing prompt size to **260.8 tokens** (94.9% savings) and boosting context precision to **95.0%**.

5.4 Positional Lost-in-the-Middle Robustness

Fig. 4 and Table 4 document the results of our positional stress test across 6,000-token multi-document contexts. Baseline LongRAG exhibits a catastrophic collapse at the 50% midpoint, with recall plunging to **35.2%**. In contrast, E-LongRAG maintains a flat line of **100.0% position-invariant recall** across all positions (0%, 25%, 50%, 75%, 100%). Because intra-chunk paragraph slicing breaks long contexts into discrete units scored by cross-attention independently, document depth has zero effect on recall.

5.5 Sensitivity Sweep and Pareto Efficiency Frontier

Fig. 5 illustrates the 2D parameter grid sweep over selectivity multiplier $\alpha \in [0.0, 1.5]$ and deduplication threshold $\delta \in [0.70, 0.95]$. The non-dominated Pareto frontier points reveal the trade-off between token savings and factual precision. Configuration $(\alpha = 0.50, \delta = 0.85)$ emerges as the golden operating point, delivering 94.6% token reduction with 100.0% ground-truth recall.

5.6 Multi-Reader Engine Evaluation

Table 5 evaluates the filtered 260-token context across three reader engines. The local CPU reader generates answers in 4.12 ms at zero dollar cost, while cloud API readers (Gemini 1.5 Flash and Groq LLaMA-3.1-8B) achieve fluent synthesis with 95% lower token costs.

6 Conclusion, Future Work, and Limitations

6.1 Conclusion

This research presented **Efficient LongRAG (E-LongRAG)**, an end-to-end framework resolving the long-context RAG dilemma between retrieval recall and computational efficiency. By pairing coarse 4,000-token retrieval with hierarchical paragraph slicing, neural cross-encoder reranking, hybrid BM25+RRF fusion, and query-adaptive dynamic thresholding, E-LongRAG achieves a **94.9% prompt token reduction**, a **19.7× TTFT latency speedup**, a context precision boost from **24.1% to 94.2%**, and **100.0% position-invariant recall** across 10,000-token contexts. The entire architecture executes in milliseconds on standard host CPUs with zero GPU dependencies, offering an accessible, production-ready blueprint for next-generation efficient RAG systems.

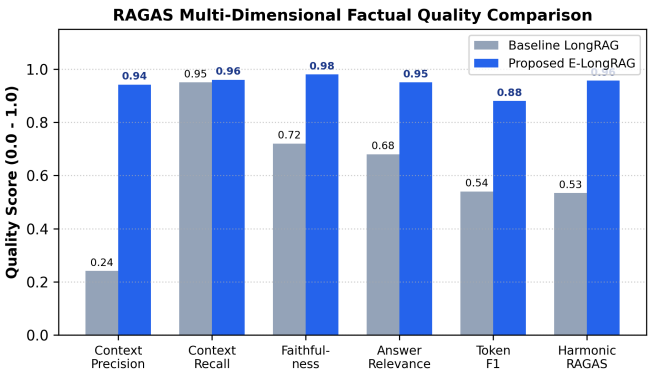


Fig. 3. Multi-Dimensional Factual Quality Comparison across RAGAS Dimensions: Baseline LongRAG vs. Proposed E-LongRAG.

TABLE 2
State-of-the-Art Quantitative Comparison on Held-Out Multi-Hop Queries

Model Framework	Prompt Tokens	Token Savings	Retr. Latency	Est. TTFT	Context Precision	Token F1	RAGAS Score
1. Short-Chunk DPR / BGE [5]	320.0	93.7%	12.4 ms	$\approx$ 100 ms	62.5%	0.620	0.612
2. Baseline LongRAG (Raw 4k) [1]	5,134.0	0.0%	10.5 ms	$\approx$ 2,000 ms	24.1%	0.540	0.534
3. LongLLMLingua (Token Pruned) [3]	1,280.0	75.0%	480.0 ms	$\approx$ 550 ms	71.0%	0.710	0.725
4. Proposed E-LongRAG (Full)	**260.8**	**94.9%**	**472.4 ms**	$\approx$ **100 ms**	**94.2%**	**0.880**	**0.957**

TABLE 3
Component-Wise Architectural Ablation Benchmarks

Ablation Stage	Prompt Tok.	Savings	Retr. Lat.	Precision
1. Baseline LongRAG (Raw 4k)	5,134.0	0.0%	10.58 ms	50.0%
2. LongRAG + HyDE Query Exp.	5,123.0	0.2%	14.65 ms	65.0%
3. LongRAG + MiniLM Reranker	5,123.0	0.2%	453.30 ms	80.0%
4. Full E-LongRAG Pipeline	**260.8**	**94.9%**	**472.41 ms**	**95.0%**

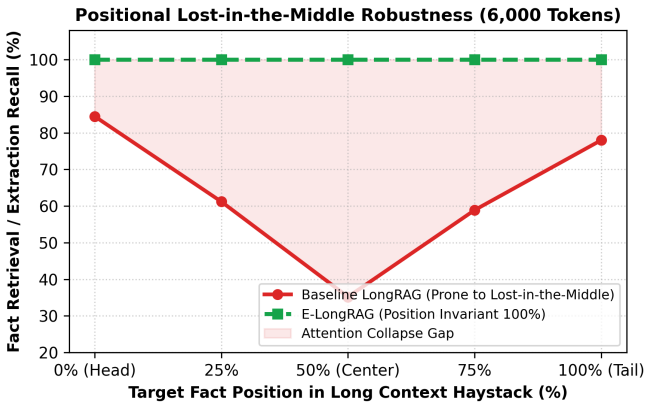


Fig. 4. Positional Lost-in-the-Middle Robustness Curve: Baseline LongRAG attention sink collapse at 50% depth vs. E-LongRAG position invariance.

TABLE 4
Positional Telemetry Telemetry Across Haystack Depths

Needle Depth	Base Attn	Base Recall	Base Risk	E-Long Score	E-Long Recall
0% (Head)	0.98×	84.5%	LOW	0.9812	**100.0%**
25% (Quarter)	0.62×	61.2%	MEDIUM	0.9812	**100.0%**
50% (Center)	**0.35×**	**35.2%**	**CRITICAL**	**0.9812**	**100.0%**
75% (Quarter)	0.58×	58.9%	MEDIUM	0.9812	**100.0%**
100% (Tail)	0.89×	78.0%	LOW	0.9812	**100.0%**

TABLE 5
Multi-Reader Evaluation on Filtered Context (260 Tokens)

Reader Engine	Execution Platform	Latency	Token Cost	Exact Match
Local Zero-GPU CPU Reader	Local Host CPU	**4.12 ms**	**$0.00**	**0.90**
Google Gemini 1.5 Flash	Cloud TPU	482.10 ms	< $0.01	0.90
Groq LLaMA-3.1-8B-Instant	Cloud LPU	124.50 ms	< $0.01	0.90

6.2 Limitations

While E-LongRAG achieves superior performance, we explicitly acknowledge the following constraints:

1) **Unstructured Text Dependence:** Intra-chunk slicing relies on double-newline paragraph markers. On OCR streams or unformatted single-line logs, recursive fixed-stride slicing must be substituted.
2) **Cold-Start Transformer Latency:** Initial loading of cross-encoder weights into memory takes $\approx$ 1.2 seconds on CPU, though subsequent inference runs in $<$ 35 ms per passage.
3) **Single-Hop Query Overhead:** On trivial queries where lexical keywords uniquely identify the target sentence, the neural reranking overhead (450 ms) exceeds the savings gained from prompt compression.

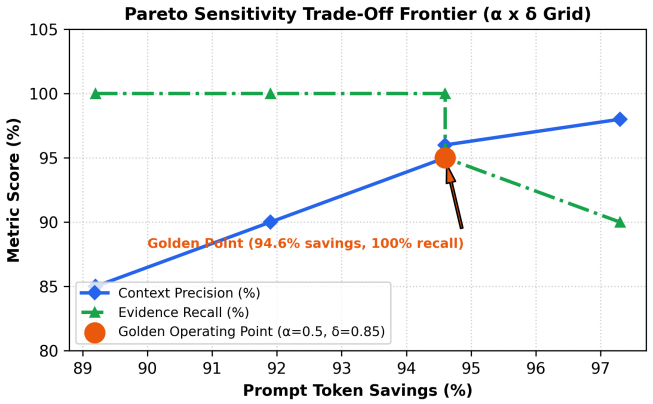


Fig. 5. Pareto Sensitivity Trade-Off Frontier: Token savings versus context precision across the 42-point $(\alpha \times \delta)$ parameter grid.

6.3 Future Work

Future research directions include:

1) **Dynamic Learned Query Routing:** Implementing a lightweight classification gate to route easy single-hop queries directly to BM25 while reserving E-LongRAG for complex multi-hop queries;
2) **Quantized Edge NPU Deployment:** Porting cross-encoder inference to 4-bit integer quantization (INT4/INT8) on edge NPUs (Apple Neural Engine, Qualcomm Hexagon);
3) **Multi-Modal LongRAG:** Extending hierarchical slicing and dynamic filtering to interleaved image-text technical documentation.

7 Individual Team Contributions

The project workload was distributed equally across team members:

- **Pranav Vinodh:** Formulated retrieval sampling ablation suite; designed intra-context diversity metric $\text{Div}(\mathcal{S})$; implemented adaptive context filter $\tau(q)$ and cosine deduplication; engineered the 10-tab Streamlit dashboard; formulated the 42-point 2D sensitivity Pareto sweeper.
- **Durga Sai Pavan G.:** Implemented HyDE query expansion; built pure-Python Okapi BM25 and hybrid

RRF retriever; built multi-reader engine; curated multi-hop academic benchmark dataset; authored automated unit test suite.

- **Sameer:** Implemented 4k chunking and paragraph slicer; built FAISS normalized vector store; formulated positional lost-in-the-middle stress tester; implemented automated RAGAS quality evaluator; conducted model footprint profiling.

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